

Poisson Distribution Examples

MDragon Data Tools - Study Guide

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1. What is a Poisson Distribution?

The **Poisson distribution** models the number of times an event occurs in a fixed interval of time or space, given that events occur:

- Independently
- At a constant average rate
- One at a time

Real-World Examples

- Number of customer arrivals per hour
- Number of emails received per day
- Number of defects per square meter of fabric
- Number of accidents at an intersection per month

- Number of calls to a call center per minute

2. When to Use Poisson Distribution

Use Poisson when:

1. **Counting occurrences** of an event in an interval
2. **Events are independent** (one doesn't affect another)
3. **Average rate (λ) is known** and constant
4. **Events cannot occur simultaneously** (very unlikely to occur at exactly the same instant)

Poisson vs Binomial

Poisson	Binomial
Number of trials unknown/very large	Fixed number of trials
Probability of success very small	Probability can be any value
Events over time/space	Fixed trials
Example: Calls per hour	Example: Coin flips

Rule of Thumb: Use Poisson when $n \geq 20$ and $p \leq 0.05$

3. Formulas and Notation

Notation

- λ (lambda) = average number of occurrences in the interval
- x = actual number of occurrences

- e = Euler's number ≈ 2.71828
- $P(X = x)$ = probability of exactly x occurrences

Poisson Probability Formula

$$P(X = x) = (\lambda^x \times e^{(-\lambda)}) / x!$$

Mean

$$\mu = \lambda$$

Standard Deviation

$$\sigma = \sqrt{\lambda}$$

Variance

$$\sigma^2 = \lambda$$

Key Insight: For Poisson distribution, the mean equals the variance!

4. Step-by-Step Examples

Example 1: Exact Probability

 **Problem:** A call center receives an average of 3 calls per minute. What is the probability of receiving exactly 5 calls in the next minute?

 **Solution:**

1. Identify the values:

- $\lambda = 3$ (average calls per minute)
- $x = 5$ (calls we want)
- $e \approx 2.71828$

2. Apply Poisson formula:

$$\begin{aligned} P(X = 5) &= (3^5 \times e^{(-3)}) / 5! \\ &= (243 \times 0.0498) / 120 \\ &= 12.10 / 120 \\ &= 0.1008 \end{aligned}$$

✓ Answer: 0.1008 or 10.08%

Example 2: At Most

 **Problem:** A hospital emergency room sees an average of 2 patients per hour. What is the probability of seeing at most 1 patient in the next hour?

 **Solution:**

1. Identify the values:

- $\lambda = 2$
- We need $P(X \leq 1) = P(0) + P(1)$

2. Calculate $P(X = 0)$:

$$\begin{aligned} P(0) &= (2^0 \times e^{(-2)}) / 0! \\ &= (1 \times 0.1353) / 1 \\ &= 0.1353 \end{aligned}$$

3. Calculate $P(X = 1)$:

$$\begin{aligned} P(1) &= (2^1 \times e^{-2}) / 1! \\ &= (2 \times 0.1353) / 1 \\ &= 0.2707 \end{aligned}$$

4. Sum the probabilities:

$$P(X \leq 1) = 0.1353 + 0.2707 = 0.4060$$

✓ Answer: 0.4060 or 40.60%

Example 3: At Least (Using Complement)

 **Problem:** A website receives an average of 5 visitors per minute. What is the probability of receiving at least 3 visitors in the next minute?

 **Solution:**

1. Identify the values:

- $\lambda = 5$
- We need $P(X \geq 3)$

2. Use complement rule:

$$\begin{aligned} P(X \geq 3) &= 1 - P(X < 3) \\ &= 1 - [P(0) + P(1) + P(2)] \end{aligned}$$

3. Calculate individual probabilities:

$$\begin{aligned} P(0) &= (5^0 \times e^{-5}) / 0! = 0.0067 \\ P(1) &= (5^1 \times e^{-5}) / 1! = 0.0337 \\ P(2) &= (5^2 \times e^{-5}) / 2! = 0.0842 \\ P(X < 3) &= 0.0067 + 0.0337 + 0.0842 = 0.1246 \end{aligned}$$

4. Apply complement:

$$P(X \geq 3) = 1 - 0.1246 = 0.8754$$

✓ Answer: 0.8754 or 87.54%

Example 4: Adjusting the Interval

 **Problem:** A bookstore sells an average of 10 books per hour. What is the probability of selling exactly 3 books in 15 minutes?

 **Solution:**

1. Adjust λ for the new interval:

- Original: 10 books per 60 minutes
- Need: books per 15 minutes

$$\begin{aligned}\lambda_{\text{new}} &= 10 \times (15/60) \\ &= 10 \times 0.25 \\ &= 2.5 \text{ books per 15 minutes}\end{aligned}$$

2. Calculate probability:

$$\begin{aligned}P(X = 3) &= (2.5^3 \times e^{-2.5}) / 3! \\ &= (15.625 \times 0.0821) / 6 \\ &= 1.283 / 6 \\ &= 0.2138\end{aligned}$$

✓ Answer: 0.2138 or 21.38%

Key Lesson: Always adjust λ to match your time interval!

Example 5: Range of Values

 **Problem:** A restaurant receives an average of 8 delivery orders per hour. What is the probability of receiving between 6 and 9 orders (inclusive) in the next hour?

 **Solution:**

1. Identify the values:

- $\lambda = 8$
- We need $P(6 \leq X \leq 9) = P(6) + P(7) + P(8) + P(9)$

2. Calculate each probability:

$$P(6) = (8^6 \times e^{-8}) / 6! = 0.1221$$

$$P(7) = (8^7 \times e^{-8}) / 7! = 0.1396$$

$$P(8) = (8^8 \times e^{-8}) / 8! = 0.1396$$

$$P(9) = (8^9 \times e^{-8}) / 9! = 0.1241$$

3. Sum the probabilities:

$$\begin{aligned} P(6 \leq X \leq 9) &= 0.1221 + 0.1396 + 0.1396 + 0.1241 \\ &= 0.5254 \end{aligned}$$

✓ **Answer:** 0.5254 or 52.54%

Example 6: Mean and Standard Deviation

 **Problem:** A highway has an average of 1.5 accidents per month. What is the standard deviation of accidents per month?

 **Solution:**

1. Identify λ :

- $\lambda = 1.5$

2. Calculate standard deviation:

$$\begin{aligned}\sigma &= \sqrt{\lambda} \\ &= \sqrt{1.5} \\ &= 1.225\end{aligned}$$

3. Calculate mean:

$$\mu = \lambda = 1.5$$

✓ Answer:

- Mean: 1.5 accidents per month
- Standard deviation: 1.225 accidents

Interpretation: The typical variation from the average is about 1.2 accidents.

Example 7: Zero Occurrences

 **Problem:** A rare bird is spotted an average of 0.5 times per week in a nature preserve. What is the probability of NOT seeing the bird in a given week?

 **Solution:**

1. Identify the values:

- $\lambda = 0.5$
- We need $P(X = 0)$

2. Calculate:

$$\begin{aligned}P(0) &= (0.5^0 \times e^{(-0.5)}) / 0! \\ &= (1 \times 0.6065) / 1 \\ &= 0.6065\end{aligned}$$

✓ Answer: 0.6065 or 60.65%

Interpretation: There's about a 61% chance of not seeing the rare bird in any given week.

5. Using the MDragon Poisson Calculator

Steps:

1. Navigate to **Calculators** → **Poisson Distribution**

2. Enter the parameters:

- **λ (lambda)**: Average rate of occurrence
- **x**: Number of events

3. Select calculation type:

- **Exactly (=)**: $P(X = x)$
- **At most ($\leq$)**: $P(X \leq x)$
- **At least ($\geq$)**: $P(X \geq x)$
- **Less than ($<$)**: $P(X < x)$
- **Greater than ($>$)**: $P(X > x)$

4. Click **Calculate** to see:

- Probability result
- Mean and standard deviation
- Visual distribution graph

Tips:

- Adjust λ to match your time interval
 - For "between" problems, use cumulative probabilities
 - Check that conditions for Poisson are met
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6. Practice Problems

Problem 1

A store receives an average of 4 customers per hour. What is the probability of receiving exactly 6 customers in the next hour?

Problem 2

A typist makes an average of 2 errors per page. What is the probability of:

- a) Exactly 0 errors on a page?
- b) At most 3 errors on a page?

Problem 3

A factory produces items at a rate where defects occur at an average of 0.8 per day. What is the probability of at least 2 defects tomorrow?

Problem 4

Phone calls arrive at a rate of 12 per hour. What is the probability of receiving exactly 3 calls in a 10-minute period?

Problem 5

A radioactive source emits particles at an average rate of 50 per minute. What is the mean and standard deviation of particle emissions per minute?

Answers to Practice Problems

Problem 1

$$\lambda = 4, x = 6$$

$$\begin{aligned} P(X = 6) &= (4^6 \times e^{-4}) / 6! \\ &= 0.1042 \text{ or } 10.42\% \end{aligned}$$

Problem 2

a) Exactly 0 errors:

$$\lambda = 2, x = 0$$

$$\begin{aligned} P(X = 0) &= (2^0 \times e^{-2}) / 0! \\ &= 0.1353 \text{ or } 13.53\% \end{aligned}$$

b) At most 3 errors:

$$\begin{aligned} P(X \leq 3) &= P(0) + P(1) + P(2) + P(3) \\ &= 0.1353 + 0.2707 + 0.2707 + 0.1804 \\ &= 0.8571 \text{ or } 85.71\% \end{aligned}$$

Problem 3

$$\lambda = 0.8, \text{ need } P(X \geq 2)$$

$$\begin{aligned} P(X \geq 2) &= 1 - [P(0) + P(1)] \\ &= 1 - [0.4493 + 0.3595] \\ &= 1 - 0.8088 \\ &= 0.1912 \text{ or } 19.12\% \end{aligned}$$

Problem 4

First adjust λ : 12 calls per 60 min = 2 calls per 10 min

$\lambda = 2$, $x = 3$

$$P(X = 3) = (2^3 \times e^{(-2)}) / 3! \\ = 0.1804 \text{ or } 18.04\%$$

Problem 5

$\lambda = 50$

$\mu = 50$ particles per minute

$\sigma = \sqrt{50} = 7.07$ particles per minute

Quick Reference

Measure	Formula
$P(X = x)$	$(\lambda^x \times e^{(-\lambda)}) / x!$
Mean (μ)	λ
Std Dev (σ)	$\sqrt{\lambda}$
Variance (σ^2)	λ

Common λ Adjustments

- Per hour → Per minute: divide by 60
- Per day → Per hour: divide by 24
- Per year → Per month: divide by 12

Use our Poisson Calculator at MDragon Data Tools for quick calculations!

MDragon Data Tools | Statistics Education Platform

mondragon-developer.github.io/statools